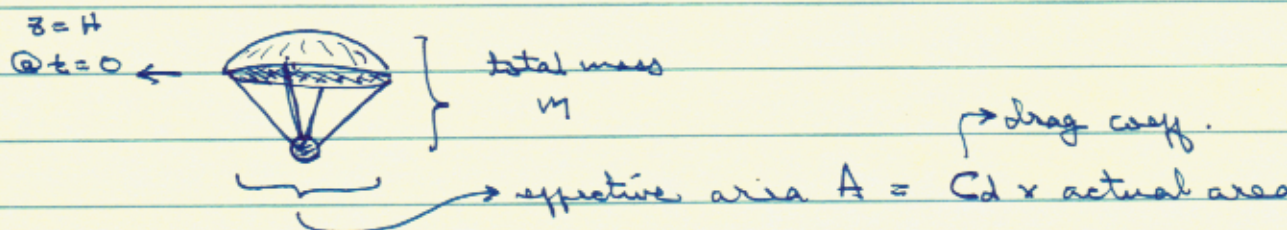


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①

Constraints on a Drop from Rest

EPP

 $z = 0$ Equation of Motion

$$\frac{du_z}{dt} = -g - \frac{A \rho_a}{2m} u_z |u_z|$$

let $L \equiv \frac{m}{A \rho_a}$, the drag attenuation length

$\Rightarrow \sqrt{gL}$ is a speed ; $\sqrt{\frac{L}{g}}$ is a time

let $u_z = (gL)^{-\frac{1}{2}} v_z$; $\tau = \left(\frac{g}{L}\right)^{\frac{1}{2}} t$
 $g = \frac{z}{L}$

$$\Rightarrow \frac{du_z}{d\tau} = -1 - \frac{1}{2} u_z |u_z|$$

multiply by u_z and note that $\frac{d}{d\tau} = \frac{d^2}{d\tau^2} \frac{d}{d\tau} = u_z \frac{d}{dz}$

$$\Rightarrow \frac{d}{d\tau} \left(\frac{1}{2} u_z^2 \right) = -1 - \left(\frac{|u_z|}{u_z} \right) \frac{1}{2} u_z^2$$

note that $|u_z|/u_z = \text{sign}(u_z) = -1$ for vertical descent

$$\Rightarrow \underline{\underline{\frac{dk}{dP} = -1 + k}} \quad ; \quad k \equiv \frac{1}{2} u_{\frac{1}{2}}^2 \text{ (specific KE)}$$

Solution in Space

$$dk (1-k)^{-1} = -dP$$

$$\Rightarrow -\ln(1-k) = -P + \text{const}$$

$P = P_0 (= H/L)$ initially, when $k=0$ (from rest)

$$\Rightarrow -\ln 1 = -P_0 + \text{const} \Rightarrow \text{const} = P_0$$

$$\Rightarrow -\ln(1-k) = P_0 - P$$

$$\Rightarrow \boxed{k(P) = 1 - e^{-(P_0 - P)}}$$

$$\text{at impact, } k = k_i = 1 - e^{-P_0}$$

when P_0 large, $k_i \rightarrow 1$, the terminal energy in these units

Solution in Time

$$k = \frac{1}{2} u_{\frac{1}{2}}^2 = \frac{1}{2} \left(\frac{dP}{dt} \right)^2 = 1 - e^{-(P_0 - P)}$$

$$\Rightarrow \frac{dP}{dt} = -2^{1/2} [1 - e^{-(P_0 - P)}]^{1/2}$$

$$\begin{aligned} \frac{dx}{dt} &= \frac{dx}{d\tau} \frac{d\tau}{dt} = \frac{dx}{d\tau} \frac{1}{\cosh x} \\ \frac{dx}{d\tau} &= \cosh x \Rightarrow \int dx = \int \cosh x dx = \ln[\cosh x + \sqrt{\cosh^2 x - 1}] \end{aligned}$$

$$\text{let } \delta = \rho_0 - \rho > 0$$

$$\Rightarrow \frac{d\delta}{d\tau} = +2^{\frac{1}{2}} [1 - e^{-\delta}]^{\frac{1}{2}}$$

$$\Rightarrow d\delta e^{\delta/2} [e^{\delta} - 1]^{-\frac{1}{2}} = 2^{\frac{1}{2}} d\tau$$

$$\text{let } \xi = e^{\delta/2} \Rightarrow d\xi = \frac{1}{2} \xi d\delta$$

$$\Rightarrow 2 d\xi [\xi^2 - 1]^{-\frac{1}{2}} = 2^{\frac{1}{2}} d\tau$$

$$\Rightarrow 2 \ln[\xi + \sqrt{\xi^2 - 1}] = 2^{\frac{1}{2}} \tau + \text{const}$$

$$\text{@ } \tau = 0, \rho = \rho_0, \delta = 0, \xi = 1$$

$$\Rightarrow 2 \ln[1] = \text{const} = 0$$

$$\Rightarrow \boxed{2^{\frac{1}{2}} \ln[\xi + \sqrt{\xi^2 - 1}] = \tau}$$

the time to impact, τ_i , corresponds to $\xi = \xi_0 = e^{\rho_0/2}$

$$\Rightarrow \boxed{2^{\frac{1}{2}} \ln[\xi_0 + \sqrt{\xi_0^2 - 1}] = \tau_i}$$

or

$$2^{\frac{1}{2}} \cosh^{-1} \xi = \tau ; 2^{\frac{1}{2}} \cosh^{-1} \xi_0 = \tau_i$$

$$\Rightarrow \xi = \cosh(2^{\frac{1}{2}} \tau) ; \xi_0 = \cosh(2^{\frac{1}{2}} \tau_i)$$

Summary in Physical Units

$$v_z^2(x) = 2gL [1 - e^{-(H-x)/L}]$$

$$v_i^2 = 2gL [1 - e^{-H/L}]$$

$$e^{(H-x)/2L} = \cosh \left[\left(\frac{g}{2L} \right)^{1/2} t \right]$$

$$e^{H/2L} = \cosh \left[\left(\frac{g}{2L} \right)^{1/2} t_i \right]$$

$$\Rightarrow t_i = \left(\frac{2L}{g} \right)^{1/2} \ln [e^{H/2L} + \sqrt{e^{H/L} - 1}]$$

Zero-Drag Limit

as $L \rightarrow \infty$,

$$v_i^2 \rightarrow 2gL [1 - 1 + H/L + \dots] \approx 2gH \quad (\text{correct})$$

$$t_i^2 \rightarrow \frac{2L}{g} \left\{ \ln \left[1 + \frac{H}{2L} + \sqrt{\frac{H}{L}} + o\left(\frac{H}{L}\right) \right] \right\}^2$$

$$= \frac{2L}{g} \left\{ \sqrt{\frac{H}{L}} + \dots \right\}^2 = \frac{2H}{g} + \dots \quad (\text{correct})$$

Numerics

$$L = \frac{m}{A \rho_a} \quad \text{at sea level} \Rightarrow \rho_a \approx 1.2 \text{ kg m}^{-3}$$

let $m_{10} = \text{m/10kg}$ and let A be in units of m^2

$$\Rightarrow \boxed{L \approx 8.3 m_{10} A^{-1} \text{ m}}$$

if $H \gg L$, then

$$\boxed{v_i^2 \approx 2gL} \quad (\text{i.e., terminal speed})$$

$$t_i \approx \left(\frac{2L}{g}\right)^{1/2} \underbrace{\ln \left[e^{\frac{H}{2L}} + e^{\frac{H}{2L}} \right]}_{= \ln 2 + \frac{H}{2L}}$$

$$\approx \left(\frac{2L}{g}\right)^{1/2} \left[\ln 2 + \frac{H}{2L} \right]$$

$$\approx \left(\frac{2L}{g}\right)^{1/2} \left(\frac{H}{2L}\right) = \boxed{\left(\frac{H^2}{2gL}\right)^{1/2}}$$

$$= \boxed{\frac{\text{distance of fall}}{\text{terminal speed}}}$$